



Final Exam_A

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Exam Instructions

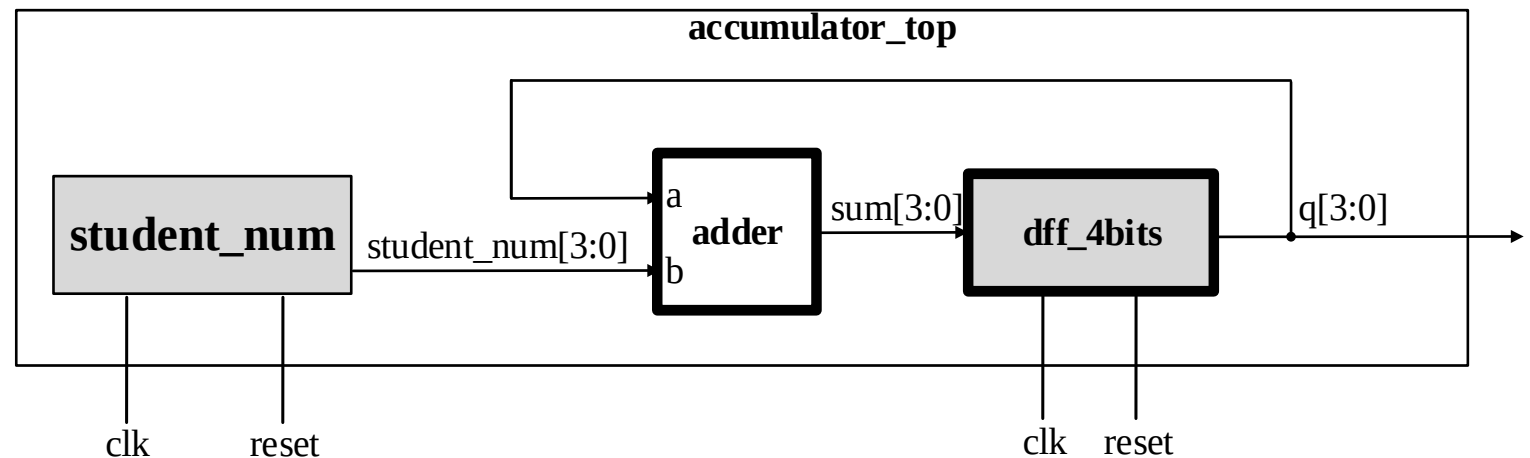
- There is only one exam question, and the grading criteria are listed at the end.
- Burning to FPGA is not required for the exam—do not use this as an excuse to walk around the classroom.
- Please copy the folder to the download before use to avoid execution errors in Quartus or ModelSim caused by Chinese characters in the file path. (You may also use the previously provided de0_empty.rar from Tronclass to write your code (not recommended).)
- The design folder inside the project directory already contains the basic .sv files you will need (the contents are empty, please write them yourself). Any additional required .sv files must be created manually.
- Hint: If you create new .sv files, be sure to update the compile.do file in simulation/modelsim accordingly.

Exam Instructions

- A testbench is provided and already placed in simulation/tb inside the project folder.
- **TAs will not assist with debugging.** Once you have completed the design and confirmed that the waveform is correct, raise your hand and ask the TA to grade your work.
- If the exam ends or you decide to give up, **rename your project folder to your student ID**, compress it into **a zip file**, and make sure the TA has copied the file before leaving.
- **If you encounter any issues during the exam, please refer to this exam instruction page first. If problems persist, raise your hand for assistance.**

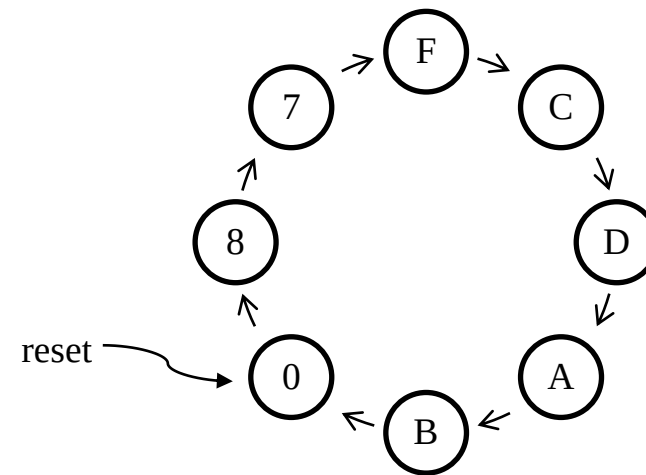
Accumulator

- Question: Design an accumulator composed of a jump-number FSM (student_num), a 4-bit adder (adder), and a 4-bit D-type flip-flop (dff_4bits). The inputs to the adder, a and b, come from the outputs of dff_4bits and student_num, respectively. The design method for student_num is explained on the next page.
- Signal settings:
 - Input: *clk* and *rst*
 - Output: *q[3:0]*
- Architecture diagram



student_num

- Output a sequence of numbers in the format FCDAB0XX, where XX represents the last two digits of your student ID. If there is a repeated digit in the sequence, replace the last occurrence of the repeated digit with E.
 - Example 1: If the student ID is 87, the output is FCDAB087
 - Example 2: If the student ID is 99, the output is FCDAB09E
- Signal setting
 - Input: *clk*, *rst*
 - Output: student_num [3:0]



- | Signal | Width | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | | | | |
|--------|-------|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|---|---|---|---|
| clk | 1 | | | | | | | | | | | | | | | | | | | | | |
| reset | 0 | | | | | | | | | | | | | | | | | | | | | |
| a | 6 | 0 | 8 | f | e | a | 7 | 1 | c | 4 | b | a | 6 | 3 | d | 8 | 0 | 7 | 6 | | | |
| b | c | 0 | 8 | 7 | f | c | d | a | b | 0 | 8 | 7 | f | c | d | a | b | 0 | 8 | 7 | f | c |
| sum | 2 | 0 | 8 | f | e | a | 7 | 1 | c | c | 4 | b | a | 6 | 3 | d | 8 | 8 | 0 | 7 | 6 | 2 |
| q | 6 | 0 | 8 | f | e | a | 7 | 1 | c | 4 | b | a | 6 | 3 | d | 8 | 0 | 7 | 6 | | | |

Grading Criteria

- The earlier you submit, the higher your score.
- If the task is not completed within the allotted time, grading will be based on the following criteria:
 - Total: 100 points, distributed as follows:
 - Karnaugh Map simplification of student_num – 40%
 - student_num code – 20%
 - adder code – 20%
 - dff_4bits code – 10%
 - accumulator_top code – 10%